

Soumyajit Das

[in](#) Soumyajit | [personal-site](#) | [✉ dsoumyajit295@gmail.com](mailto:dsoumyajit295@gmail.com) | [+91 8538852773](#)

OBJECTIVE

A physics graduate with a keen interest in semiconductor physics, condensed matter physics, and computational materials science. I am interested in first-principles simulations, nanoscience, MEMS/NEMS, and machine learning applications in physics. I want to contribute towards the scientific community through my skills and learning.

EDUCATION

National Institute of Technology Silchar

M.Sc. in Physics; CGPA: 8.46

2024 – 2026

Silchar, Assam, India

University of Calcutta(Asutosh College)

B.Sc. (Honours) in Physics; CGPA: 8.11/10

2020 – 2023

Kolkata, ,West Bengal, India

West Bengal Council of Higher Secondary Education

Higher Secondary Examination: Percentage: 96%

2020

West Bengal, India

West Bengal Board of Secondary Education

Secondary Examination; Percentage:93%

2018

West Bengal, India

RESEARCH INTERESTS

Condensed Matter Physics, DFT and Computational Materials Science, Superconductivity, Nanoscience, Machine Learning Applications in Physics.

RESEARCH EXPERIENCE

Summer Research Intern, IISER Kolkata

2025

Worked on first-principles DFT calculations using VASP and Elk; performed structural relaxations, band structure and electronic property analysis of crystalline materials.

PROJECTS

M.Sc Project:*Ab-initio DFT Study of Monolayer and Bilayer ZrS₂/ZrSe₂*

- Performed first-principles DFT calculations using VASP to study electronic and optical properties of 2D TMDCs.
- Investigated SOC, DOS/PDOS, band structure, optical conductivity, and stacking effects.
- Studied the influence of layer thickness (monolayer vs bilayer) and AA/AB stacking on electronic properties and bandgap engineering.

LABORATORY AND TECHNICAL SKILLS

Scientific Computing: VASP, ELK, VESTA, Python, NumPy, Matplotlib, PyQt6, L^AT_EX

Computational Methods: Density Functional Theory (DFT), PBE-GGA, PAW, Band Structure, DOS/PDOS, Spin-Orbit Coupling (SOC), Optical Properties

Operating Systems Windows, Linux (Ubuntu, Linux Mint, CentOS)

CONFERENCES, WEBINARS AND WORKSHOPS

- Participated in the webinar on **Pattern Formation in Nature**, presented by Prof. Jayanta Kumar Bhattacharjee, Distinguished Visiting Professor, School of Physical Sciences, IACS Kolkata. **2021**
- Participated in the webinar on **The Story of our Star the Sun**, presented by Prof. Dibyendu Nandi, Department of Physical Sciences, IISER Kolkata. **2021**
- Achieved **1st rank** in the **One-Day Workshop on Applied Electronics**, organised by the Department of Physics, Asutosh College, University of Calcutta. **2022**

CERTIFICATIONS

- Two-Day National Webinar on **Fundamental Physics**, Asutosh College. 2021
- **1st Position** — One-Day Workshop on **Applied Electronics**, Asutosh College. 2022
- Basic Computer Course and Web Designing — **Grade A**, Asutosh College. 2021

ACHIEVEMENTS

- **IIT JAM 2024**: Secured **All India Rank (AIR) 1436**.
- Secured **1st position** among **40 students** in the **One-Day Workshop on Applied Electronics**.
- Selected for and successfully completed a **Summer Research Internship** at **IISER Kolkata**.
- **Higher Secondary Examination**: Scored **480 out of 500**.

EXTRACURRICULAR ACTIVITIES

- Independent learning of web developement, software developement
- Science and technology blogging
- Photography
- Drawing